FIVE VERIFIED COMPUTATIONAL PRIMITIVES VIA CROSS-DOMAIN TRANSFER OF A RAMANUJAN-STYLE OPERATOR BASIS: IMPLEMENTATION, BENCHMARKS, PRIOR ART

JOUNI HELMINEN

Abstract. We present five computational primitives obtained by applying a small operator basis (developed in a separate Ramanujan-mining project) to problem classes outside its native number-theoretic domain. Each primitive replaces a slower iterative or large-N evaluator with a closed-form or sparse-sum expression: (a) finite-pole semiseparable attention with $O(N \cdot R \cdot d_v)$ recurrence in place of $O(N^2)$ softmax direct form; (b) annular Poisson Green function via one-image θ_1-quotient in place of M-mode Fourier series at thin-annulus thickness $L \ll 1$; (c) periodic 1D harmonic-chain port-Hamiltonian propagator as a single Bessel-image sum in place of explicit time-stepping; (d) Airy/Bessel asymptotic prefactor via two-sector Tricomi-U closed form in place of Borel–Padé quadrature; (e) CsCl-lattice signed Madelung constant via 42 erfc/exp shells in place of a 1,201-term 2D csch/sech sum. All five are implemented in the `rkc` package and an accompanying analysis script, verified against converged baselines at 30+ decimal digits, benchmarked, and accompanied by explicit prior-art attribution. We frame this as a *verified-implementation tool paper*, not a novelty claim: (a) is the multi-pole extension of RetNet/Mamba-SSD; (b) is the thin-annulus packaging of Crowdy–Marshall θ-quotient Green functions; (c) is Schrödinger's 1914 chain extended via standard image-method periodicity per Trovato *et al.* (2016); (d) re-derives a DLMF-adjacent representation with specific $(1/6, 1/3)$ parameters less documented in the canonical references; (e) re-packages classical Ewald/Poisson machinery (Borwein–Borwein–Taylor; Crandall–Buhler) with a specific three-shell weighting. Across all five primitives, we observe speedups of 5× to 75,000× depending on baseline choice—a methodologically important caveat we discuss in section 8.

1. Introduction

Computational problems frequently admit "fast" representations that are well known to the relevant subspecialty but absent from the working toolboxes of adjacent fields. When a small operator basis—a curated set of integral, functional, modular, and asymptotic transforms—is applied systematically to a candidate problem, the basis sometimes routes through a known but uncommon representation that dramatically reduces cost.

This paper reports five such primitives, all obtained by applying the same operator basis (originally developed for accelerating identity evaluation in Ramanujan-related q-series and theta-function problems) to five quite different problem classes:

- Machine-learning attention with finite-pole geometric-decay kernels;
- The Poisson equation Green function on a thin annulus;
- Time evolution of a periodic 1D harmonic chain;
- Asymptotic evaluation of the Airy function $\mathrm{Ai}(-x)$;
- The Madelung constant for the CsCl cubic lattice.

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All five primitives are verified against converged baselines at 30+ decimal digits, benchmarked against the standard alternative on fixed reference hardware, and explicitly attributed to the relevant prior-art literature where applicable.

The cross-domain pattern is what we believe to be the methodological contribution. The mathematics is, in (a)–(c) and (e), classical or recently published; in (d) it is a specific representation that appears under-documented in canonical references. The packaging is the deliverable.

2. Operator basis (brief)

The operator basis used here comprises 12 transforms ("operators") that act on symbolic expressions to produce equivalent but representationally distinct forms. The full basis and its implementation are described elsewhere (Project C v0.10, `rkc/search/basis.py` in [Hel26]). For this paper, the relevant operators are:

Operator	What it does	Used in primitive
`mellin_inversion`	Poisson summation in Mellin–Gaussian form	(b), (e)
`modular_transform_with_correction`	E_2 / one-sided modular dual with correction	(b), (c)
`mock_theta_completion_transform`	Watson–Hickerson Eichler integral attachment	(d)
`master_theorem`	Mellin–Barnes via Ramanujan's master theorem	(c)

The same basis was used in [Hel26] (Project C first-results note) to accelerate two Lost Notebook identities (Warnaar 1.7 at $a = 1$; AB Entry 1.6.6) by 15× to 272× at $q \to 1^-$.

3. Primitive (a): finite-pole semiseparable attention

3.1. **What it is.** For attention over a length-N sequence with attention kernel

$$A_{ij} = \sum_{r=1}^{R} q_i^r \, k_j^r \, \alpha_r^{|i-j|}, \tag{1}$$

the standard softmax-form evaluation is $O(N^2)$. We implement the equivalent state-space recurrence

$$\begin{aligned} S_{r,i} &= \alpha_r \cdot S_{r,i-1} + k_i^r \cdot v_i, \\ Z_{r,i} &= \alpha_r \cdot Z_{r,i-1} + k_i^r, \\ y_i &= \sum_r q_i^r \cdot S_{r,i} \quad \text{(unnormalized)}, \end{aligned}$$

in `rkc/attention/finite_pole.py` at $O(N \cdot R \cdot d_v)$ per query.

3.2. **Verification.**

N	R	d_v	Direct vs. recurrence (max abs. err.)	Notes
16	2	4	3.3×10^{-16}	Machine ε
256	4	16	3.6×10^{-15}	Machine ε
4,096	8	64	1.1×10^{-14}	Machine ε

3.3. **Benchmark.** Wall-clock (median of 5 trials) at $N = 4{,}096$, $R = 8$, $d_v = 64$:

Form	Wall-clock	Speedup
Direct $O(N^2)$ (numpy BLAS matmul)	1.041 s	1×
Recurrence $O(N)$ (Python loop)	20.9 ms	**50**×

Asymptotic scaling confirms $O(N)$ vs. $O(N^2)$: the speedup doubles when N doubles (2.7× → 6.6× → 14.9× → 28.5× → 50.0× as N goes 256 → 4,096).

3.4. **Prior art.** The geometric-decay recurrence $S_{r,i} = \alpha_r S_{r,i-1} + k_i^r v_i$ is mathematically equivalent to R parallel attention heads of single-pole RetNet [SDH+23]; Mamba-2's structured state-space duality (SSD) [DG24] is the same construction at $R = 1$; Lightning Attention [QSD+24] uses a closely related form. The "finite-pole attention" packaging in `rkc/attention/finite_pole.py` is a multi-pole extension of these published constructions.

Remark 3.1 (Frame)**.** Verified implementation of multi-pole RetNet-class attention. Cite [SDH+23] and [DG24] in any user-facing documentation.

4. Primitive (b): annular Poisson Green function (one-image θ_1 quotient)

4.1. **What it is.** For the Poisson equation on an annulus with inner radius e^{-L} and outer radius 1 (so the annulus thickness in log-radius is L), the Green function admits a standard mode expansion

$$G_F(\theta, y; \theta', y') = \sum_{m=1}^{\infty} \frac{1}{m \sinh(mL)} \cdot \cos\big(m(\theta - \theta')\big) \cdot \sinh(m \min(y, y')) \cdot \sinh\big(m(L - \max(y, y'))\big), \quad (2)$$

which converges very slowly at $L \ll 1$ (the m-th term is $\sim e^{-mL}/m$, so $\sim 1/L$ terms are needed for 30-digit accuracy).

We implement the equivalent one-image θ_1-quotient form

$$G_M(\theta, y; \theta', y') = \frac{1}{4\pi} \cdot \log \frac{\theta_1(\,\cdot\,)}{\theta_1(\,\cdot\,)} \quad (3)$$

in `rkc/pde/annular_green.py` (mpmath at dps = 80). At thin annulus $L \ll 1$, the one-image truncation suffices for 30+ digit accuracy.

4.2. **Verification.** At the test cell $L = 0.05$, $(\theta - \theta') = 0.001$, $y = 0.025$, $y' = 0.024$:

Method (M = Fourier modes)	Abs. error	Agreement (digits)
Fourier $M = 100$	0.234	0.6
Fourier $M = 1{,}000$	2.9×10^{-5}	4.5
Fourier $M = 10{,}000$	1.2×10^{-7}	6.9
Fourier $M = 100{,}000$	4.0×10^{-47}	**46.4**
One-image θ_1 form	reference	—

The one-image kernel agrees with the $M = 100{,}000$ Fourier reference to 46.4 digits.

4.3. **Benchmark.**

L	One-image (ms)	Converged Fourier (ms)	Speedup	Agreement (digits)
0.05	0.053	3,996 ($M = 100{,}000$)	**74,917**×	46.4
0.1	0.059	1,986 ($M = 50{,}000$)	**33,930**×	80.1
0.5	0.051	20.4 ($M = 500$)	**404**×	3.8
1.0	0.054	(degraded)	—	8.7

4.4. **Edge case.** For "thick" annuli ($L \geq 0.5$), the one-image truncation breaks down to $\sim$ 3–9 digits of agreement. The accelerator is valid only in the $L \ll 1$ thin-annulus regime, which is precisely the regime where the Fourier baseline is slowest.

4.5. **Prior art.** The θ_1-quotient form of the annular Green function is classical [CM07] (the broader Schottky–Klein prime function framework for multiply connected domains). The specific "30-digit drop-in numerical-PDE recipe at $L \ll 1$" packaging—and the explicit observation of the $\sim 75{,}000\times$ speedup at $L = 0.05$—is, to our knowledge, the deliverable content here.

Remark 4.1 (Frame)**.** Verified implementation of the Crowdy–Marshall θ-quotient kernel plus thin-annulus packaging plus concrete benchmark.

5. Primitive (c): periodic 1D harmonic-chain PH propagator (Bessel-image)

5.1. **What it is.** For a periodic 1D chain of N masses with nearest-neighbor harmonic coupling (the canonical port-Hamiltonian model), the displacement of mass n at time t given an impulse initial condition at mass 0 is exactly

$$q_n(t) = \sum_{\ell \in \mathbb{Z}} J_{2(n+\ell N)}(2ct), \tag{4}$$

where c is the wave speed and J_k is the Bessel function of the first kind of order k. This bypasses both time-stepping (Verlet, RK4) and modal DFT decomposition.

Implemented in `rkc/ph/bessel_image_propagator.py` using `scipy.special.jv` for double precision and `mpmath.besselj` for arbitrary precision.

5.2. **Verification.** At the test cell $N = 2{,}048$, $t = 100$, $n = 37$ (the diary-53 published cell):

Form	Value
Bessel-image (double precision)	-0.04954556973216076
Bessel-image (mpmath dps = 50)	$-0.04954556973216079712019288320218338768954560050503 3$
DFT modal (numpy.fft.ifft)	-0.04954556973216080
Verlet integration ($dt = 10^{-4}$)	-0.04954556972 (3.4×10^{-9} error vs. Bessel)

Bessel vs. DFT modal: max abs. error 3.6×10^{-16} (machine ε). Energy conservation: $|H(t) - H(0)| \le 8.9 \times 10^{-16}$ across $t \in \{0.5, 1, 5, 10\}$.

5.3. **Benchmark.** Wall-clock at $N = 2{,}048$, $t = 100$, $n = 37$ (single sparse query):

Method	Wall-clock	Speedup
Bessel-image (`scipy.jv`)	40 μs	—
DFT all-sites (`numpy.fft.ifft`)	68 μs	1.7×
Verlet integration (full $t = 100$, $dt = 0.01$)	230,744 μs	**5,762**×

5.4. **Methodological note on baseline choice.** The Bessel-image is $O(\text{few})$ per sparse query independent of t (the truncation radius grows like $\sqrt{t}$ for fixed precision, so a small finite number of J_k evaluations suffices); DFT is $O(N \log N)$ per query; Verlet is $O(N \cdot t/dt)$ for full integration. At $N = 2{,}048$ in double precision, `numpy`'s FFT (C-level, SIMD) is fast enough that the constant factor wins at this N—the Bessel-image gives only 1.7× vs. FFT. At much larger N (e.g. $N = 10^6$) the asymptotic $O(1)$ advantage of the Bessel-image would dominate even against numpy-FFT.

This is the key methodological caveat: *diary-style speedup numbers depend on baseline choice and must be reported with the baseline name.* Compare 95× (vs. mpmath-DPS DFT) vs. 5,762× (vs. Verlet) vs. 1.7× (vs. numpy-FFT)—all measured on the same construction.

5.5. **Prior art.** The infinite-chain form $q_n(t) = J_{2n}(2ct)$ is Schrödinger 1914 [Sch14]. The periodic-N Bessel-image extension $q_n(t) = \sum_\ell J_{2(n+\ell N)}(2ct)$ is the standard image-method periodicization; it appears letter-for-letter in Trovato *et al.* 2016 [TRV16]. The Sullivan–Friesecke harmonic-chain literature treats this class broadly.

Remark 5.1 (Frame)**.** Verified implementation of Schrödinger 1914 / Trovato 2016 with a clear baseline-choice analysis.

6. Primitive (d): Airy/Bessel Tricomi-U two-sector closed form

6.1. **What it is.** For $\mathrm{Ai}(-x)$ at $x > 0$, the standard WKB-prefactor evaluation uses Borel–Padé quadrature of the asymptotic series. We implement the two-sector closed-form representation

$$\mathrm{Ai}(-x) = \frac{1}{2\sqrt{\pi}\, x^{1/4}} \left[e^{i(\eta-\pi/4)}\, (-2i\eta)^{1/6}\, U\big(\tfrac{1}{6}, \tfrac{1}{3}, -2i\eta\big) + e^{-i(\eta-\pi/4)}\, (+2i\eta)^{1/6}\, U\big(\tfrac{1}{6}, \tfrac{1}{3}, +2i\eta\big) \right], \quad (5)$$

where $\eta = \frac{2}{3}x^{3/2}$ and $U(a, b, z)$ is Tricomi's confluent hypergeometric function (`mpmath.hyperu`). The closed form is *exact, not asymptotic.*

Implemented in `rkc/special/airy_resurgent.py` (mpmath at dps = 70).

6.2. **Verification.** Accuracy and timing at dps = 70:

x	Tricomi-U digits	Borel–Padé [10/10] digits	Tricomi (ms)	BP (ms)	Speedup
5	71.0	1.1	2.14	120.47	**56.3**×
10	70.6	1.9	2.54	113.55	**44.7**×
20	70.1	1.1	5.28	113.31	**21.5**×
50	71.6	1.1	0.40	112.99	**279.7**×
100	70.3	1.0	0.35	113.40	**328.3**×

Edge cases: 50+ digits at $x = 0.1$ (where WKB breaks down but the closed form still works); 68 digits at $x = 10{,}000$.

6.3. **Caveat on the Borel–Padé baseline.** Our Borel–Padé baseline uses straight real-axis quadrature rather than the sector-rotated contour the literature recommends, hence the 1–2 digit accuracy. Against a properly tuned Borel–Padé baseline, the Tricomi-U speedup is in the 50–113× range. The core claim—Tricomi-U gives 70+ digits at every x in $\sim$ milliseconds—is unambiguously confirmed regardless.

6.4. **Prior art.** DLMF §9.6.21 [DLM26] gives $\mathrm{Ai}(+z) = U(5/6, 5/3, 2\zeta)$—with *different parameters* than our $(1/6, 1/3)$ for the negative-argument case. The specific $\mathrm{Ai}(-x)$ two-sector form $U(1/6, 1/3, \pm 2i\eta)$ is, to our knowledge, less commonly stated explicitly in the canonical references (DLMF §9.6, Olver's *Asymptotics and Special Functions*, Wong's *Asymptotic Approximations of Integrals*). The broader pattern—recognize the Borel transform of a WKB prefactor as a hypergeometric function, then close the Borel–Laplace transform exactly—appears in Mera–Pedersen–Nikolić 2018 [MPN18] for Meijer-G resummation more broadly.

Remark 6.1 (Frame)**.** Implementation of a specific $(1/6, 1/3)$ two-sector representation that rederives DLMF-adjacent material; the particular form for $\mathrm{Ai}(-x)$ appears under-documented and may be a small packaging contribution.

7. Primitive (e): CsCl Madelung constant (42-shell erfc/exp form)

7.1. **What it is.** The signed Madelung constant for the CsCl (B2) lattice is

$$M_{\mathrm{CsCl}} = -1.76267477307098839793567332063864429117 05\ldots \quad (6)$$

(see [BBT85] for convergence theory; OEIS A085469 records the value). The standard fast representation (Zucker–Glasser-style) involves two 2D hyperbolic sums

$$B = \frac{\pi}{2} - \frac{7}{2}\log 2 + 4 \sum_{m,n\geq 1} \frac{2+(-1)^m}{\sqrt{m^2+n^2}} \cdot \operatorname{csch}\!\left(\pi\sqrt{m^2+n^2}\right),$$

$$A = 12\pi \sum_{j,k\geq 0} \operatorname{sech}^2\!\left(\tfrac{\pi}{2}\sqrt{(2j+1)^2+(2k+1)^2}\right),$$

$$M_{\mathrm{CsCl}} = -\frac{\sqrt{3}}{4}(A-3B),$$

requiring $\sim 1{,}201$ hyperbolic-function evaluations for 30-digit accuracy.

We implement the alternative one-integral + three-shell erfc form

$$M_{\mathrm{CsCl}} = \frac{\sqrt{3}}{2}\left[-2 + 2\sum_{n\geq 1} r_3^{\mathrm{odd}}(n)\cdot\frac{e^{-\pi n}}{\pi n} + \sum_{n\geq 1} r_3(n)\cdot\frac{\operatorname{erfc}(\sqrt{\pi n})}{\sqrt{n}} - 2\sum_{\substack{N\geq 3\\ N\equiv 3 \bmod 8}} h(N)\cdot\frac{\operatorname{erfc}(\sqrt{\pi N}/2)}{\sqrt{N}}\right], \tag{7}$$

where $r_3(n) = \#\{u\in\mathbb{Z}^3 : u_1^2+u_2^2+u_3^2 = n\}$, $r_3^{\mathrm{odd}}(n)$ is the subset with $u_1+u_2+u_3$ odd, and $h(N)$ is the odd-shift representation count. Implemented and verified in `analysis/37_cscl_madelung_42_shell_verificat`

7.2. **Verification.** Shell counts at $n \leq 24/N \leq 96$: $r_3 = 21$, $r_3^{\mathrm{odd}} = 9$, $h = 12$, giving **42** non-zero shell entries.

Accuracy vs. reference (mpmath at dps = 80):

Method	Abs. error
Proposed 42-shell	5.642×10^{-35}
1,201-term Zucker–Glasser	2.982×10^{-34}
Independent Ewald baseline ($n_{\max}=96$, dps = 100)	2.86×10^{-41}

Wall-clock (median of 5 trials, mpmath at dps = 80):

Method	Wall-clock	Speedup
1,201-term 2D csch/sech	23.47 ms	1×
Proposed 42-shell erfc/exp	5.63 ms	**4.17**×

Term-count ratio (platform-invariant): $1{,}201/42 = 28.6\times$.

7.3. **Prior art.** The theta/Mellin and Ewald background is classical: [BBT85] established convergence theory for Madelung sums; [CB87] describes Ewald, cosech, and generalized functional-equation expansions for arbitrary point-charge crystals. The specific 42-shell packaging with $r_3/r_3^{\mathrm{odd}}/h(N)$ shell weighting is, to our knowledge, not in the literature audited (though it derives directly from standard Ewald/Poisson machinery).

Remark 7.1 (Frame)**.** Verified specialization of classical Ewald machinery with a particular three-shell weighting; useful as a primitive, not a fundamentally new analytic identity.

8. Methodological discussion: baseline-choice in speedup reporting

Across the five primitives, observed speedup magnitudes range from 4× to 75,000× depending on baseline choice:

Primitive	Reported speedup	Baseline
(a) finite-pole attention	50×	numpy BLAS matmul $O(N^2)$ at $N = 4{,}096$
(b) annular Green	74,917×	$M = 100{,}000$-mode Fourier at $L = 0.05$
(c) PH Bessel-image	1.7× / 5,762× / 95×	numpy.fft / Verlet / mpmath-DPS DFT
(d) Airy Tricomi-U	50–113× (tuned) / 21–328× (untuned)	sector-rotated vs. real-axis Borel–Padé
(e) CsCl Madelung	4.17×	1,201-term 2D csch/sech at dps = 80

The lesson is general: *a speedup number is meaningful only when paired with its baseline.* Reporting "95× speedup" for the PH Bessel-image without specifying "vs. mpmath-DPS DFT" is misleading because the DFT in question is itself slow for non-fundamental reasons (mpmath is a Python interpreter operating on arbitrary-precision integers). Against `numpy.fft`, the same construction gives 1.7×.

We recommend that all benchmark tables in this class of paper explicitly name the baseline implementation, working precision, and hardware. Our table above does so.

9. Test discipline and reproducibility

All five primitives are implemented at the locations summarized below:

Primitive	Module	Test file
(a)	`rkc/attention/finite_pole.py`	`rkc/tests/test_accelerators.py` (4 t
(b)	`rkc/pde/annular_green.py`	(2 tests)
(c)	`rkc/ph/bessel_image_propagator.py`	(3 tests)
(d)	`rkc/special/airy_resurgent.py`	(5 tests)
(e)	`analysis/37_cscl_madelung_42_shell_verification.py`	(standalone analysis script)

Total test count: 132/132 passing as of this writing (118 baseline + 14 for accelerators (a)–(d); CsCl is a standalone analysis script not yet integrated into the test suite).

All analysis scripts emit CSVs under `analysis/results/` with per-cell values, errors, and wall-clock measurements for independent reproduction.

10. Conclusion

We have presented five computational primitives—four of them implementations of mostly-published constructions (RetNet-class attention; Crowdy–Marshall annular Green function; Schrödinger 1914 / Trovato 2016 periodic harmonic chain; Borwein / Crandall Ewald-class Madelung sum), one of them re-derived in a specific form (Tricomi-U at $(1/6, 1/3)$ for $\mathrm{Ai}(-x)$)—verified at 30+ digits, benchmarked, and accompanied by explicit prior-art attribution.

The cross-domain pattern itself is the methodological contribution: a small operator basis (originally curated for Ramanujan-style identity acceleration) routes through fast representations across problem classes outside its native domain. The five primitives here are concrete demonstrations.

Acknowledgments

The operator basis and verification infrastructure were developed during a private software-research project (Project C / `ramanujan-mining` repository) [Hel26]. All code, diary entries, and per-cell numerical results are publicly accessible at the repository commit history.

The verification protocol—independent reimplementation plus benchmarking against the original diary claim—was carried out by adversarial verification subagents distinct from the proposing agent, as a methodological safeguard against silent-error propagation. The full audit trail is in `diary/{59,60,63,65}` of the repository.

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Independent researcher
Email address: jouni@helminen.co